



Flying Liferaft
Wing Structural Design



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Abstract

The report outlines the structural design of the wings on a light UAS capable of VTOL. A flight envelope, V-n diagram, was constructed according to FAR 23 for a utility aircraft. The ultimate load factor was used to size the skin thickness and spar dimensions. No stringers or ribs were required due to the low loads observed on the wings.

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1 Introduction

Maritime rescue operations are often time slow and expensive to carryout. The project aimed to design a UAS capable of carrying and deploying modular payloads. Research was done into the type of payload possible for rescuing drowning people and a liferaft and buoys that met the payload weight were found. To meet the design specifications of no overhead equipment, the UAS has a VTOL system mounted on the wings [1]. There were two load cases considered: the VTOL case and maximum load factor during a maneuver case. The more constraining of the two would be used to design the structural components.

2 Structural Layout Considerations

In the conceptual design, the wings were placed above the payload area to not shift the center of gravity too much after the deployment of the payloads. The additional unused space above the payload lends itself to a wing box carry through design. The structure would not interfere with other systems such as the opening of the "bomb" bay doors. The fuselage heavy frames would connect to the two spars of the wing box to transfer the loads from the wings to the frames [2]. The spar locations were chosen from historical data on typical spar locations and shown in Table 1.

Component	Location (% of Chord)
Front Spar	15
Rear Spar	70

Table 1: Spar Locations

The Aerodynamic team provided the sizing of the planform of the wings. The pylon span-wise location was chosen to be 0.85 m from the root [3]. The rear spar location as well as the pylon positioning was then pass on to Performance, Stability, and Control to size the ailerons. Figure 1 showcases the wing planform and the spar locations.

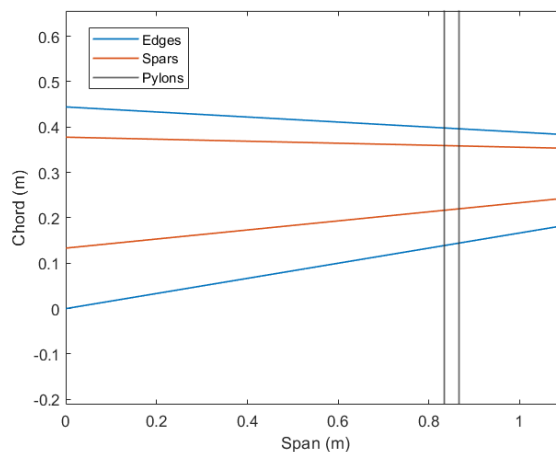


Figure 1: Top view of the wing planform, spar locations, and pylon location

As for the connection of the spars to the pylons, a stepped-lap joint design to join the composite pylon to a hollow cylinder of aluminium alloy for welding was considered. The stepped-lap joint would provide the best shear strength, but the complexity of manufacturing such a complex joint at such a small size would prove infeasible. Instead, the pylons would be slotted through holes drilled in both spars. The cutout would have a lip to alleviate the stress concentration. Then a L-shaped piece of aluminium would be joined to the pylon using epoxy/silicon and bolted to the spar.

3 Load Cases

3.1 Maneuver and Gust

The maneuvering and gust loads were considered using a Vn diagram following the FAR 23 utility category aircraft. The maneuver load factors of +3 and -1 were deemed a suitable option from Raymer and Gundlach for ISR drones and strategic bombers [4] [5]. Both types of aircraft have a similar mission profile to the aircraft being designed. These values allow for good maneuverability without imposing an unnecessarily large constraint on the structure of the aircraft. Figure 2 shows the Vn diagram developed and Table 2 shows the key values on the Vn diagram.

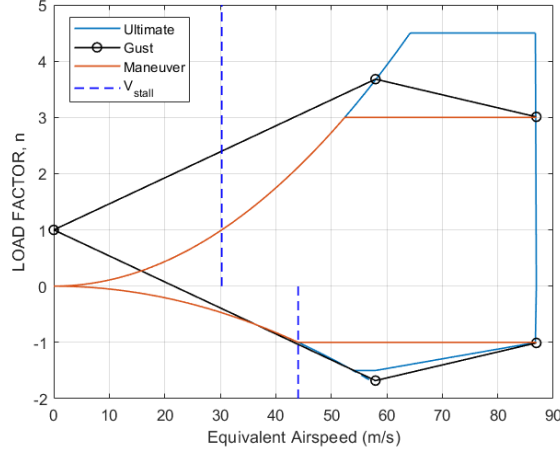


Figure 2: Vn diagram

Maneuver		
Case	EAS (m/s)	n
Stall	30.4 / 44.0	1.0 / -1.0
Cruise	58.0	3.6 / -1.5
Dive	86.9	4.5 / -1.0
Gust		
Case	EAS (m/s)	n
Stall	-	-
Cruise	58.0	3.6 / -1.7
Dive	86.9	4.5 / -1.0

Table 2: Key Values from Vn Diagram

Although the drone would not need to be certified under FAR 23, some of the requirements from FAR 23 were used in the design anyway. These requirements would give a good reference for what good handling and typical airspeeds the drone should be able to reach at different stages of flight. The minimum dive speed for a utility category aircraft is defined as $1.5V_c$, where V_c is the equivalent airspeed at cruise. Discussions with the Performance, Stability, and Control team showed that the selected dive speed would be more than enough to complete the payload deployment maneuver [6]. The design speed for maximum gust intensity, V_b would not be considered and taken as V_c as the drone would not have any passengers, thus passenger or pilot comfort do not need to be taken into consideration.

The vertical gust speeds, U_{de} , used to calculate the gust loads on the Vn diagram are shown in Table 3. These values are rather large as the cruise height laid well below the 20,000 feet mark where

the vertical gust during different stages begin to get smaller. However, two considerations had to be made in regards to the gust intensity seen by the UAV. Firstly, the UAV is much lighter than other aircrafts certified under FAR 23. This would increase the effect of the gust on the UAV. Secondly, the intensity of the gust follows a cosine shape rather than a step function as the aircraft penetrates through the gust. This reduces the intensity of the gust. To account for both factors, Equation 1 gives a more accurate value for vertical gusts experience by the UAV using a correction factor [5].

$$U = KU_{de} \quad (1)$$

Where

K:

$$K = \frac{0.88\mu}{5.3 + \mu} = 0.826 \quad (2)$$

Mass ratio:

$$\mu = \frac{2(W/S)}{\rho g \bar{c} a} \quad (3)$$

Where W/S is the wing loading, $\bar{c}$ is the mean aerodynamic chord, and a is the lift curve slope in radians. The results can be seen in Table 3. Overall, the correction factor reduces the gust speed by around 20%. The corrected vertical gust speed was then used to calculate the change in load factor using Equation 4.

$$\Delta n = \frac{\rho U V a}{2(W/S)} \quad (4)$$

Where V is the equivalent airspeed of the aircraft. The gust loads are applied to the UAS during steady level flight where $n = 1$ and for both vertical gusts acting upwards and downwards.

Cases	$U_{de}(fps)$	U (fps)	Δn
Cruise	50	41.3	2.68
Dive	25	20.7	2.01

Table 3: Caption

The Vn diagram shows that the UAV would be on the verge of stalling from a 50 feet per second gust during cruise. The value of the $C_{L,max}$ used for the plot was 1.26 which is the lower value of the $C_{L,max}$ estimates from the Aerodynamics team [9]. A higher value for $C_{L,max}$ would improve this issue. Overall, the +4.5 maneuver load case was the most constraining for the maneuver and gust load case and will be used in comparison to the VTOL case.

3.2 VTOL

During this stage of the flight, ideally the wings would not be generating lift. The loads on the wings instead come from the motors attached to the wings with pylons. Each of the VTOL motors will generate 151 N of thrust according to the Propulsion and Systems team [7]. The front pylon would transfer the load into the front spar and it is the same respectfully for the rear pylon. Figure 3 shows the connection between the wings and the pylons.

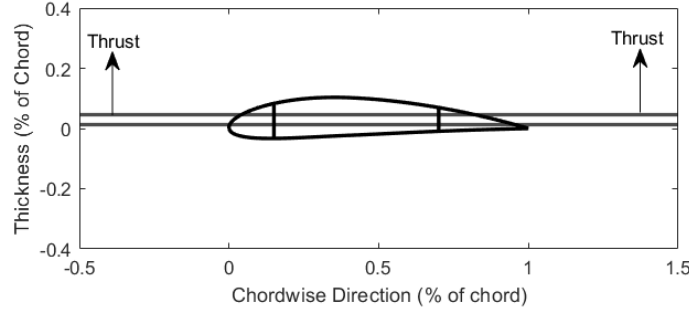


Figure 3: Side View of the Wing and Pylons connection

However, these conditions are ideal which would not be truly representative of the actual operation of the UAS. The drone is required to be able to operate in all weather conditions, which would include the case where there are strong gales. These gales will create aerodynamic loads in addition to the loads from the motor thrusts. A safety factor will need to be considered.

3.3 Bending Moment, Shear Force, and Torque

In the cases described above, the shear force, bending moment, and torque were calculated using the classical beam theory, where the wing is modelled as a cantilever. The initial total wing mass estimation, M_w , was done so using Equation 5 [8].

$$M_w = 0.03(N_z \times MTOW)^{0.826} \times AR^{0.28} \times S_w^{0.19} \times (1 + \lambda)^{0.21} \times t/c^{0.14} = 3.51kg \quad (5)$$

Where N_z is the ultimate load factor, $MTOW$ is the maximum takeoff weight in kilograms, AR is the aspect ratio, S_w is the wing area, λ is the taper ratio, and t/c is the thickness to chord ratio. The equation gave a very small initial estimate because there were not enough data points covering the range of $MTOW$ the UAV currently sits at for the empirical equation to be accurate. However, this only served as a starting point for future iterations to obtain more accurate values for the wing mass. The wing mass was then distributed along the span using Equation 6.

$$\omega = \frac{2 \times 9.81 \times M_w}{b(1 + \lambda)} \left(1 - \frac{1 - \lambda}{b/2} z\right) \quad (6)$$

Where b is the span, and z is the distance along the span starting at 0 at the root. The distribution is an idealized trapezoidal distribution assuming $\omega_{max} = \lambda \times \omega_{min}$. The actual mass of each section can be calculated directly from the structures sized from previous iterations. The pylons and motors on each side of the wing weighs 352 and 1260 grams respectfully [3]. These values are added to the distribution at the span location of the pylon which is located at $z = 0.85m$. This inertial distribution was then used for both load cases.

3.3.1 Maneuver Case

In the +4.5 maneuver case, the data points for lift distribution along the span during cruise was obtained from the Aerodynamic team [9]. The values were then multiplied by the load factor and integrated numerically using trapezoid integration along the span to obtain the shear force contribution. The numerical integration was completed once again to obtain the bending moment contribution

from lift. These were combined with inertial contributions to get the total shear force and bending moment. The results are plotted in Figure 4,5, and 6.

To obtain the torque distribution some assumptions were made. The flexural axis of the wing box was assumed to be at the center of the wings box. The lift force was assumed to act through the quarter-chord point. This assumption holds well as the airfoil chosen is quite thin and the UAV is flying at low speeds. In reality, this assumption may well be a very slight overestimate as the aerodynamic center tends to shift towards the trailing edge which reduces the moment arm. The center of gravity of a section of the wings was calculated from the center of gravity of the skin and spars. The distance between the center of gravity and the flexural axis was used as the moment arm for inertial loads.

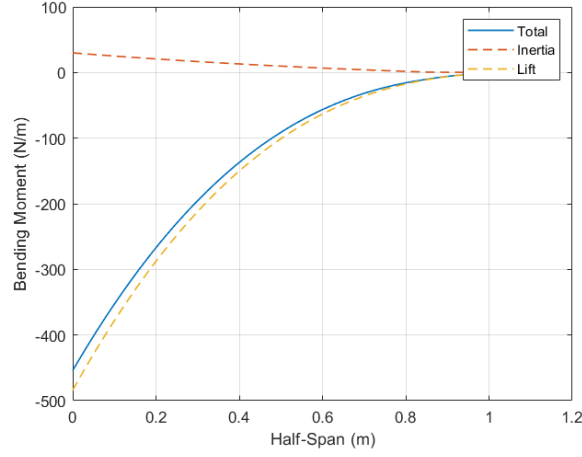


Figure 4: Bending Moment along the span due to maneuver

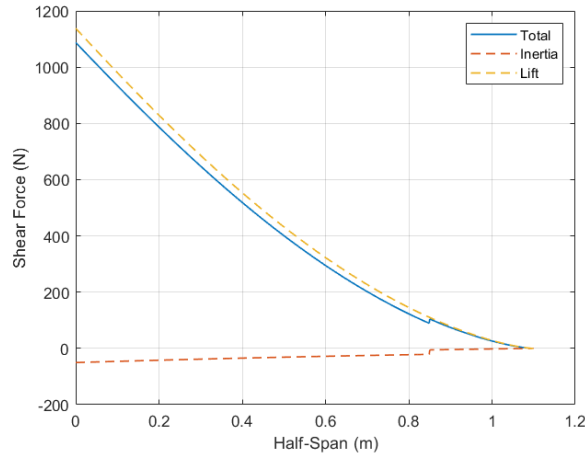


Figure 5: Shear Force along the span due to maneuver

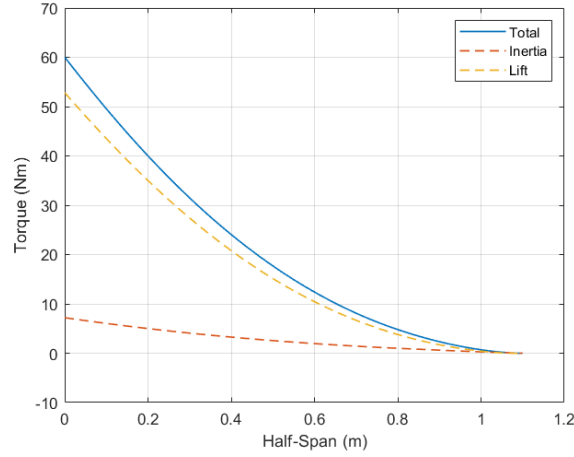


Figure 6: Torque along the span due to maneuver

In Figure 4 and 5, the inertial contribution reduces the magnitude of both the bending moment and shear force, which allows the structure to be lighter theoretically. However, the center of gravity of the structures (i.e. spars and skin) was located behind the flexural axis; thus, the torque of lift and weight add to each other. The presence of the pylons and motors can be seen at around $0.85m$ along the span. All of the graphs reach 0 at the tip of the wing giving validation that the calculations are correct. All the graphs should reach 0 at the tip because the wing is modeled as a cantilever with the tip being the free end.

3.3.2 VTOL

The forces on the pylons were translated to the wings with the shear force acting as a point load and the bending moment were added as torque contributions. Figure 7,8, and 9 shows the bending moment, shear force, and torque that were generated from a vertical take-off maneuver.

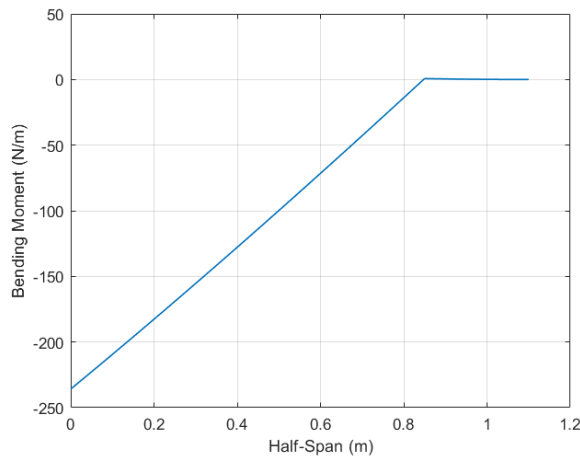


Figure 7: Bending Moment along the span due to VTOL

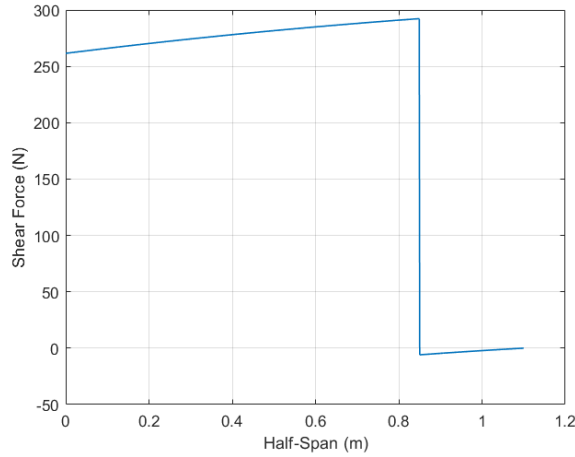


Figure 8: Shear Force along the span due to VTOL

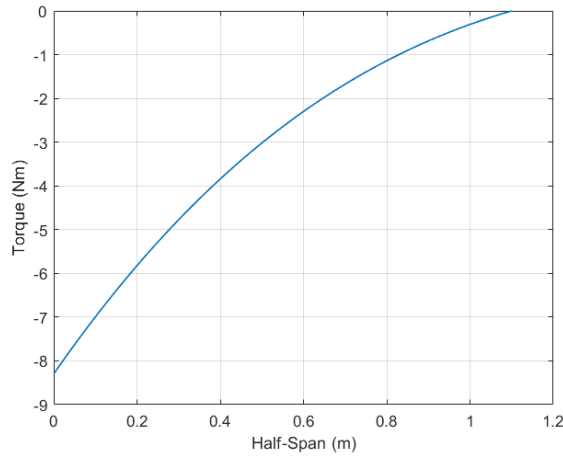


Figure 9: Torque along the span due to VTOL

The two main assumption in generating these plots were that the wings generate no lift at all and the drag force from moving the wing vertically upwards is negligible. The calculations are still valid as from inspection those effects should cancel each other out, however, further calculations into this was not done. The note worthy feature on the graphs are the maximum magnitude on each plot. In comparison to Figure 4, the maximum magnitude in Figure 7 is almost 2 times less. Figure 8 and 9 are similarly smaller than the maneuver case. The torque was not as high as expected at the attachment point of the front and rear pylon. This is because the rear pylon is only slight longer than the front pylon and most of the torque from the pylons on the wings gets cancelled out at that location. Another concern was the slope due to the deflection of the pylon would vector the thrust towards the center of the wings creating a larger compressive force as well. This case was taken into consideration during the sizing of the structural components as well for comparison.

4 Structural Component Design

4.1 Spars

The spars resists the torque and the shear loads. The torque and shear loads were converted to shear stress, τ , using the Bred-Batho formula shown in Equation 7.

$$\tau = \frac{q}{t} = \frac{T}{2At} = \frac{F}{2ht} \quad (7)$$

Where q is the shear flow, T is the torque, F is the shear load, A is the area of the wing box, t is the thickness, and h is the height of the wing box. The shear flow in the front spar and rear spar are defined in Equation 8 and 9.

$$q_{FS} = \frac{F}{2ht} + \frac{T}{2At} \quad (8)$$

$$q_{RS} = \frac{F}{2ht} - \frac{T}{2At} \quad (9)$$

For structural efficiency, the spars would ideally buckle under the shear flow calculated. Equation 10 provides the relation between the shear flow and thickness of the spar webs [10].

$$\frac{q}{t} = 8.1E\left(\frac{t}{h}\right)^2 \quad (10)$$

The resulting thicknesses can be seen in Figure 10.

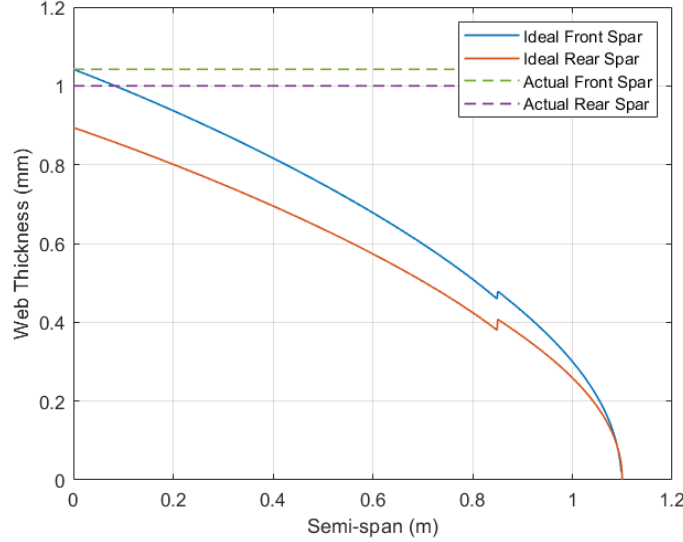


Figure 10: Ideal and Actual Skin Thickness distribution along the span

As can be seen, the spar web thickness are below the manufacturing limits of aluminium. The two main reasons for keeping the two spar design. Firstly, the one spar design would still require a false spar to be the attachment point for ailerons and the rear spar were already as thin as they could be. The second reason being the second spar also provides a redundancy in the load path. With a slightly questionable joint between the spar and the pylons, the design would benefit from having two points to transfer loads to the frames.

4.2 Skin

To maximize the efficiency of the structure, the skin thickness were designed to buckle at the ultimate load factor selected. The relation between the skin thickness and the buckling stress, σ , can be seen in Equation 11 [10].

$$\sigma = \frac{N}{t} = 3.62E\left(\frac{t}{w}\right)^2 \quad (11)$$

Where N is the compressive load per unit length, and w is the distance between the front and rear spars. N can be found using the bending moment obtained from the +4.5 maneuver case and dividing by the area of the wing box. The resultant thicknesses along the span is shown in Figure 11.

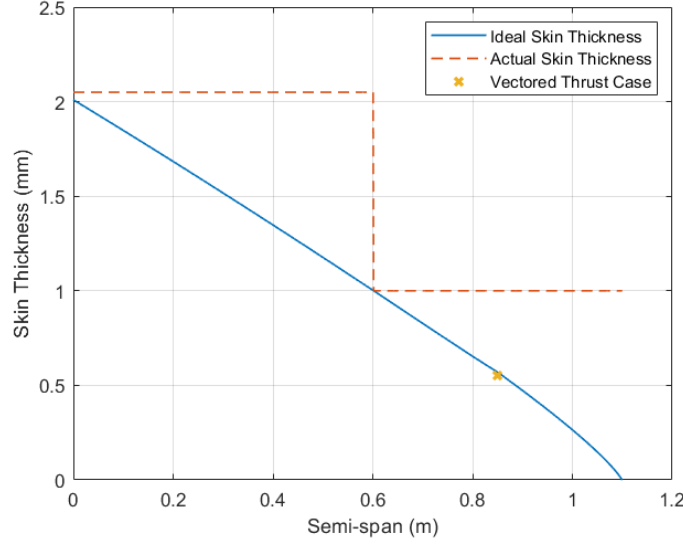


Figure 11: Ideal and Actual Spar Thickness distribution along the span

The orange cross on the plot represents the thickness of the ribs required to withstand buckling under the vectored thrust due to beam deflection case. Due to manufacturing constraints the ideal thickness can not be achieved for over half of the semi-span. The actual skin thickness could be varied more often to better match the ideal distribution, but having multiple small sectors with different skin thickness should be avoided unless really necessary. It was found that varying the skin thickness once would allow the drone to meet the MTOW set by the group [11]. To address the fact that the skin would not only be carrying compressive load but also shear flow around the wing box, combined loading analysis was done. Equation 12 shows the maximum combined loading at the root section [10].

$$R_s^2 + R_c = 0.961 \leq 0.99 \quad (12)$$

Where R_s is the shear stress ratio, and R_c is the compressive stress ratio. The calculated value is below 0.99 indicating the skin would survive the combined loading.

4.3 Ribs

The ribs were designed to withstand the crush load from bending moments and maintain the aerodynamic shape. The crush load, F_{crush} , and rib thickness, t_r , can be calculated using Equation 13 [10].

$$\frac{F_{crush}}{t_r c} = \frac{M^2 L h t_{skin} c}{2 E I^2} = 3.62 E \left(\frac{t}{h}\right)^2 \quad (13)$$

Where M is the bending moment, L is the rib spacing, t_{skin} is the thickness of the skin, c is the chord. The ideal rib thickness calculated was plotted in ??.

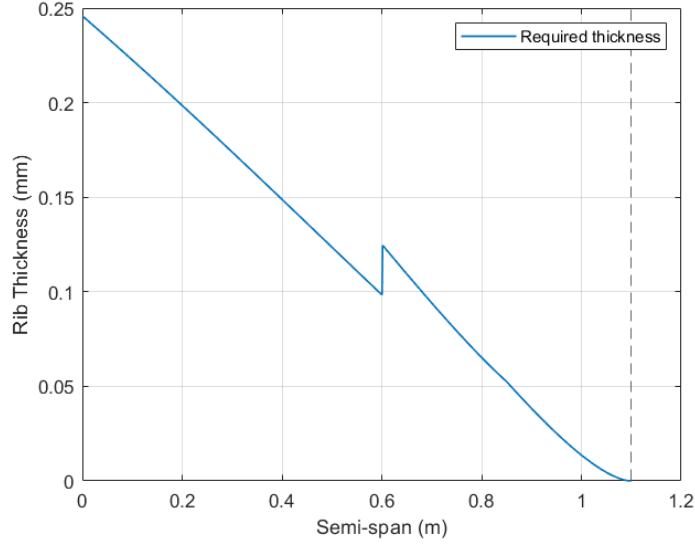


Figure 12: Ideal and Actual Rib Thickness distribution along the span

From the requirement of the ribs buckling at the maximum load case, the rib thickness become tiny. The conclusion drawn from these calculations is that there is no need for ribs to withstand the crush load. However, one rib and pseudo-rib will be placed at the tip and another set at the root. This is to provide points to rivet the skin to maintain the airfoil shape. All of these structures will be at 1 mm thickness.

5 Conclusions

The summary of the total mass of the whole wing can be seen in Table 4. The overall design was not as efficient as the team would hope to achieve. To improve on the mass reduction further, another material needed to be considered. However, due to being exposed to salty and wet environments, the corrosion resistance limited the pool of material available. Thus, the structure remains quite heavy and over designed. The trade off between complexity of manufacture and mass reduction was often made to meet the *MTOW* requirement as well as potentially saving cost during manufacturing. Another important trade off was the ones between different FATTs. During the final iteration of the wing planform, the Aerodynamic team increased the span of the wings for a higher *AR*. The result was an increase of almost 1 kg on the wings.

Components	Ideal Mass (kg)	Actual Mass (kg)	Inefficiency (%)
Skin	4.96	6.94	39.8
Spar	0.272	0.405	49.0
Ribs	0.0246	0.130	430
Total	5.26	7.47	42.1

Table 4: Summary of the Wing Design Results

References

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Appendix